INCH-POUND

MIL-R-49203(CR)
AMENDMENT 2
31 January 1989
SUPERSEDING
AMENDMENT 1
11 April 1988

MILITARY SPECIFICATION

RECORDER-REPRODUCER SET, SOUND AN/TNH-23( )

This amendment forms a part of Military Specification MIL-R-49203(CR), dated 4 March 1980, and is approved for use within the Communications-Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Preamble: Delete "Communications Research and Development Command" and substitute "Communications-Electronics Command".

DD 1426 Notice: Line 3, delete" U.S. Army Communications Research and Development Command, ATTN: DRDCO-CM-DM" and substitute "U.S. Army Communications-Electronics Command, ATTN: AMSEL-ED-TO".

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2.1 Under Specifications, Military, delete the following:
" MIL-P-116 Preservation-packaging, Methods of "

2.1. Under Standards, Military, delete the following:
" MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment Subsystems and Systems

MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of "

3.1 Replace second sentence with the following:
"It shall consist basically of either two(2) shipping/carrying/storage cases and two AC powered single deck recorder/playback units or one(1) shipping/carrying/storage case and one AC powered dual deck recorder/playback unit, one battery powered recorder/playback unit, a control changeover switch unit(for 2 single deck recorder configuration only), microphones, headset and a supply of blank cassettes."

3.1 Delete last sentence and substitute "The system shall consist of an assembly of items which interface to provide the working system described in detail herein." .

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3.2 Line 1, after "order" add "(see 6.2c)".

3.3.1 Line 1 and line 2, change the word "units" to "unit(s)".

3.3.1 Line 3, delete "height shall not be greater than one third the width".

3.3.2 Delete line 1 and substitute: " Cabinetry of the recorder/playback unit(s) (and transcriber (if required)) shall be construc-"

3.4 Add the following subparagraph:
"o. Transcriber unit,one each(if required)."

3.4.a Delete and substitute:
"a. Single deck recorder/playback unit,AC powered,2 each or dual deck recorder/playback unit,AC powered, 1 each."

3.4.f After "each" add "(for dual deck recorder, switchover control shall be internal)." .

3.4.i Delete paragraph.

3.4.j After "storage" change to read "(2 for single deck recorder configuration or 1 for dual deck recorder configuration and if transcriber is required, then 1 for the transcriber)." .

3.4.k After "adapter," insert "for battery powered recorder/playback unit,".

3.4.m Change to read as follows:
" m. Microphone "omnidirectional" type, four each."

3.4.n Delete paragraph and substitute:
"n. 115 ± 5 volt/220 ± 10 volt plug adapter, three each."

3.4.1 Delete the first three lines and substitute " Each AN/TNH-23 shall include either 2 single deck recorder/playback units or 1 dual deck recorder/playback unit that meet the specific requirements that apply to these recorders. The dual deck recorders shall record on either two or four tracks(single pass) and shall have a discrete track playback capability. The AC powered recorder/" .

3.4.1 Line 4 and 9, change "units" to "unit(s)".

3.4.1 Line 5, delete "117 volt" and substitute "115 ± 5 volt" and delete "220 volt" and substitute "220 ± 10 volt".

3.4.1 Line 5, after "cords" insert"(3 wire with inclusion of Grnd wire or 2 wire for double insulated machines)".

3.4.1 Line 8, delete "117/220V" and substitute "115 ± 5V/220 ± 10V" .

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3.4.2 Delete third and fourth sentences and substitute "This unit shall contain batteries which shall be capable of operating the recorder for at least two hours in the record mode. The unit shall be capable of using zinc alkaline batteries BA-3058/U. The unit shall be capable of operating from a 115±5V/220±10V source when used with the AC adapter. An external microphone shall be supplied with this unit and when in use this external microphone shall disconnect the internal microphone." .

3.4.3 Line 3, delete the word "three".

4.4 Line 2, change ""reverse"" to ""rewind"".

4.4 Delete "4.4" and substitute "3.4.4".

3.4.5 Line 3, after "powered" insert "single deck".

3.4.5 Line 4, after "unit." insert the sentence "For the dual deck recorder configuration, the microphone shall be usable with either deck." .

3.4.5 Line 5, change ""reverse"" to ""rewind"".

3.4.6 Line 1, insert at the beginning of the sentence "For the 2 single deck recorder configuration".

3.4.6 Line 9, change ""reverse"" to ""rewind"".

3.4.6 Line 10, delete "open-room" and substitute "omnidirectional".

3.4.6 Line 10, after "3.4.12." add the following sentence:
"For the dual deck recorder configuration the switchover control unit shall be internal and shall automatically switch to the other tape when the first ape is at or near the end. Also all audio inputs and control functions of the hand microphone shall be transferred automatically."

3.4.7 Delete "such as the "Stenomask" as manufactured by Talk Inc., 455 Union Ave., Westbury, Long Island, N.Y., or equal".

3.4.7 Line 6, change line to read "either the switchover control unit(2 single deck recorder configuration) or the dual deck recorder itself and function properly." .

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3.4.8 Line 3, replace "on each side of the cassette" with "single pass".

3.4.9 Delete paragraph and substitute:
"3.4.9 Shipping/Carrying/Storage Case(s). Ruggedized cases with removable top covers shall be furnished. No tools shall be required to remove/install the covers. The equipment inside the case(s) shall be protected from moisture, dirt and shock."

3.4.10 Delete line 2 and 3 and substitute "The adapter shall operate from 115 ± 5 Vac, 60 Hz and 220 ± 10 Vac, 50 Hz power lines. The output of the adapter shall plug directly into the battery powered recorder/reproducer unit and shall automatically disconnect the internal batteries." .

3.4.12 Line 1, in heading delete "("Open-Room" Type)" and substitute "("Omnidirectional" Type)", and in the first sentence delete ""open-room"" and substitute ""omnidirectional"".

3.4.12 Line 5, delete "Stenomask" and substitute "privacy".

3.4.12 Line 5, after "microphone." add the sentence "For the dual deck recorder configuration the microphone shall plug into the recorder itself." .

3.4.12 After paragraph 3.4.12 add paragraph "3.4.13" as follows:

"3.4.13 Transcriber Unit. If the tape recorder(s) supplied cannot perform the transcription functions outlined in this specification, a separate transcriber unit must be supplied. This unit must be compatible with the tape speed and format the recorder(s) utilizes and must meet the requirements of sections 3.3.1, 3.3.2,electrical rqmts of 3.4.1, 3.4.9, 3.4.11, 3.5, 3.6, 3.7, 3.8.1, 3.8.2, 3.8.5, 3.8.7, 3.9.1.1, 3.9.1.2, 3.9.4, 3.9.6, 3.10, 3.12 and 3.13."

3.8.3 Line 1, after "unit" insert "(single deck recorder configuration)".

3.8.3 Line 6, after "cassette" add "For the dual deck recorder configuration an "end of tape" signal is not required for automatic changeover. However, a "no-cassette alarm" is required since during initial startup or automatic changeover the activated deck may not contain a cassette(see 4.9.3)." .

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3.8.5 Delete and substitute:

"3.8.5 Recording Head. The heads of the recorder reproducer unit(s) shall meet the requirements of 4.9.5."

3.8.6 Delete and substitute:

"3.8.6 Rewind Time. Total rewind time of the one-pass cassette shall not exceed 240 seconds for both AC powered and battery powered recorder/playback units."

3.8.7 Line 1, after "be" insert " either 15/16 or".

3.8.7 Line 3, after "units" change the sentence to read "or transcribers by at least ± 15%." .

3.8.8 Delete paragraph.

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3.9.3 Delete paragraph.

3.9.4 Delete and substitute:

"3.9.4 Frequency Response. The frequency response of the AC powered and the battery powered recorder/playback units (through the record/playback cycle) shall be within ± 5 dB from 250 to 5000Hz."

3.9.5 Delete paragraph.

3.11 Delete paragraph.

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4.2c.(2) Delete in its entirety.

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TABLE I, under "Functional", change "Recording Head Position" to read "Recording Head".

TABLE I, under "Electrical", delete "Signal-to-Noise Ratio 3.9.5 4.10.5".

TABLE II, under "Functional", delete "WOW and Flutter 3.8.8 4.9.8".

TABLE II, under "Electrical", delete "Bias Frequencies 3.9.3 4.10.3".

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TABLE III, subgroup II, delete entire subgroup.

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4.7 Line 2, change "3.4.12" to "3.4.13".

4.7 After paragraph 4.7 add paragraph "4.7.1" as follows:

"4.7.1 Transcriber. If a separate transcriber unit is required, inspect, examine and operate the transcriber to determine compliance with 3.4.13."

4.9.3b. Line 1, replace "AC powered recorder units" with "cassette decks".

4.9.5 Lines 1 and 2, In heading delete "Position" and delete first sentence.

4.9.5 Line 4, delete "vertically".

4.9.6 Line 2, after "compliance" insert "of each recorder deck".

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4.9.6c Delete third sentence and substitute "Stop tape movement when stop watch indicates 240 seconds for both AC powered and battery powered recorder/playback units".

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4.9.7 Lines 3 and 4, delete second sentence.

4.9.7 Last sentence, after the word "speed" change to read "from the normal 1 7/8" or 15/16" per second shall be performed as follows:" .

4.9.7d Line 1, delete "1250Hz" and substitute "1150Hz".

4.9.7d Line 2, delete "750Hz" and substitute "850Hz".

4.9.7d Line 4, delete "±25%" and substitute "±15%".

4.9.8 Delete, including subparagraphs.

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4.10.1.1 Line 4, delete "333 Hz" and substitute "1000 Hz".

4.10.1.1 Line 8, after "(VTVM)" add "or equal".

4.10.1.1c Line 1, delete "333 Hz" and substitute "1000 Hz".

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4.10.1.2c Line 1, delete "333 Hz" and substitute "1000 Hz".

4.10.2a Line 1, after "VTVM" add "or equal".

4.10.2c and 4.10.2d Line 1, delete "333 Hz" and substitute "1000 Hz".

4.10.3 Delete, including subparagraphs.

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4.10.4a Line 2, after "VTVM" add "or equal".

4.10.4d Line 1, delete "333 Hz" and substitute "1000 Hz".

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4.10.4e Lines 1 and 2, change to read "Record a 250, 1000 (repeat of reference), 2000, 3000, and a 5000 Hz tone" and in line 4 delete "7000 and".

4.10.4f Line 2, delete "333 Hz" and substitute "1000 Hz".

4.10.4h Delete and substitute:
"h. The response through the recorder/playback cycle for both AC powered and battery powered units shall be within ± 5 dB of that of the reference tone between 250 and 5000 Hz."

4.10.5 Delete including subparagraphs.

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4.10.6b and 4.10.6c Line 1, delete "333 Hz" and substitute "1000 Hz".

4.11.1 Delete and substitute:

"4.11.1 Humidity. The equipment in its transit case shall be tested in accordance with Method 507.2, Procedure II of MIL-STD-810."

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4.11.2 Delete and substitute:

"4.11.2 Bounce, Loose Cargo. The equipment in its transit case shall be tested in accordance with Method 514.3, Procedure II of MIL-STD-810."

4.11.3 Line 2, delete "516.2, Procedure II" and substitute "Method 516.3, Procedure IV ".

4.11.4 Line 2, delete "506.1" and substitute "506.2".

4.12 Delete paragraph.

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4.13.c Line 2, delete "220 volt" and substitute "220 ± 10 volt".

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4.15 Delete paragraph.

5.1 Delete paragraph and substitute the following:
" 5.1 Packaging Requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity(see 6.2). "

6.1 Line 3, replace "the same equipment" with "this system".

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6.1 Line 6, delete "117/220 volt" and substitute "117 ± 5 volt/220 ±10 volt".

6.2b. Delete and substitute with the following:
" b. Levels of preservation and packing(see section 5). "

6.3 Delete paragraph.

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